

Website Architecture & Technical Specification Document

Target Audience: Engineering & Development Team

Platform Scope: Pan-India Hackathons, Events, Corporate Services, and Incubation Platform

Integration Partner: Devnovate Registration Gateway

1. Executive Summary & Objectives

The objective of this project is to develop a responsive, high-performance web platform designed to showcase our nationwide offerings, manage event listings, facilitate corporate and college inquiries, and route event participants smoothly to external registration pipelines (Devnovate).

Key Functional Requirements

- **Core Navigation:** Home, Events, Our Services, Our Team, Contact.
- **Events Ecosystem:**
 - User authentication (Login / Registration).
 - Push notification subscription system.
 - Dynamic event discovery with category filtering.
 - External event registration routing to the Devnovate registration portal.
 - Scroll-triggered popups for event promotions and sponsor advertisements.
- **Comprehensive Services Showcase:** Detailed nationwide delivery of 11 core services including Hackathons, Startup Incubation, Funding, Corporate Placement Drives, Goodies, and Sponsorship.
- **Lead Capture & Inquiries:** Role-based inquiry collection for colleges, corporate partners, and sponsors.

2. Information Architecture & Routing Structure

Route Breakdown

- [/](#) — **Home Page**
 - Hero banner with dynamic CTA.
 - Snapshot of services and pan-India reach ticker.
 - Featured upcoming events carousel.
 - Testimonials, past partner logos, and community metrics.
- [/events](#) — **Events Directory**
 - Category filters (Hackathons, Corporate Challenges, Placement Drives, Webinars).
 - Search and location toggle (Pan-India / Virtual / City-specific).
 - Web Push Notification opt-in prompt.
 - Scroll-triggered ad popup modal (configured at 50% scroll depth).
 - Event Cards with external registration redirect triggers.
- [/services](#) — **Our Services Directory**
 - Detailed service cards for all 11 business offerings.
 - Instant lead capture modal ("Request This Service").
- [/our-team](#) — **Team & Advisors**
 - Leadership bios, core operational team, and advisory network.
- [/contact](#) — **Contact & Business Inquiries**
 - Contact details, inquiry form with service-type dropdown, and campus ambassador outreach.
- [/auth/*](#) — **Authentication Flow**
 - [/login](#), [/register](#), [/forgot-password](#), [/profile](#).

3. Recommended Technology Stack

Layer	Technology	Rationale
Frontend Framework	Next.js (React 18/19)	Server-Side Rendering (SSR) & Static Site

Layer	Technology	Rationale
		Generation (SSG) for high SEO ranking and fast loading.
Styling & UI	Tailwind CSS + Radix UI / Shadcn UI	Responsive, accessible, customizable component architecture.
Backend & APIs	Node.js (Next.js API Routes or Express.js)	Unified TypeScript codebase, lightweight serverless deployment.
Database	PostgreSQL (hosted on Supabase or Neon)	ACID-compliant relational model ideal for users, event meta, and leads.
ORM / Query Builder	Prisma ORM	Type-safe database queries and automated schema migrations.
Authentication	NextAuth.js / Supabase Auth	Secure session cookies, email magic links, and Google OAuth support.
Push Notifications	Firebase Cloud Messaging (FCM) / OneSignal	Full Web Push API compliance across mobile and desktop browsers.
Media & Assets	AWS S3 or Cloudinary	Optimized CDN delivery for ad banners, event graphics, and merchandise assets.

Layer	Technology	Rationale
Hosting & CI/CD	Vercel	Seamless edge deployments, zero-downtime rollouts, and built-in analytics.

4. Key Functional Specifications

4.1. Event Registration & Devnovate Redirection

- The "Register" button on any event card or detail page must **not** collect internal payments or form submissions.
- Clicking the button must track conversion analytics and open the Devnovate registration URL in a new tab.
- **Outbound Link Logic:**
 - Append UTM parameters for conversion attribution (e.g., `?utm_source=platform&utm_medium=events_page&utm_campaign=hackathon_2026`).
 - Security parameters: `rel="noopener noreferrer"`.

4.2. Scroll-Triggered Ad Popups

- **Trigger Rule:** When a user scrolls past 50% of the `/events` page height.
- **Frequency Capping:** Store a timestamp in `sessionStorage` or `localStorage` to ensure the ad appears at most **once per session** per visitor.
- **Dismissal Controls:** Close button (X), clicking outside the modal overlay, or pressing the `Escape` key.
- **Ad Creative Schema:** Admin-managed banner image, headline, and outbound target URL.

4.3. Browser Push Notification Architecture

- **Opt-in Prompt:** Trigger a non-intrusive banner on the [/events](#) page: "Get notified about upcoming hackathons & placement drives."
- **Token Handling:** On grant, register the browser Service Worker, receive the FCM/OneSignal token, and POST to [/api/notifications/subscribe](#).
- **Targeting:** Capability to broadcast to all registered devices or filter by interests (e.g., Hackathons only).

5. Services Directory Specification (Pan-India)

The platform must provide dedicated visual cards and inquiry mechanisms for the following 11 core offerings:

1. **Hackathons:** End-to-end hackathon management, problem statement curation, judging platforms, and live support.
2. **Corporate Hackathons Organization:** Bespoke innovation and hiring challenges tailored for tech enterprises.
3. **Digital Marketing:** Pan-India campus outreach, digital community amplification, and targeted student marketing.
4. **Organizing Events:** Turnkey technical event management, logistics, audio-visual setup, and execution.
5. **Startup Incubation:** Mentorship programs, prototype acceleration, and pitch readiness.
6. **Providing Funding:** Connecting vetted startup ideas with angel investors, seed grants, and venture networks.
7. **Mentors Providing from Working Professionals:** On-demand access to senior engineers and corporate mentors.
8. **Internships:** Direct placement pipelines connecting hackathon performers with verified startup internships.
9. **Sponsorship:** Facilitating brand sponsorships and monetary partnerships for student clubs and colleges.
10. **Goodies & Swag:** Custom event t-shirts, badges, stickers, and kit production with pan-India delivery.

11. **Placement Drives:** Comprehensive recruitment drives, screening tests, and campus placement drives across colleges.

6. Database Schema Design (PostgreSQL DDL)

-- 1. Users Table

```
CREATE TABLE users (  
  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    full_name VARCHAR(120) NOT NULL,  
  
    email VARCHAR(255) UNIQUE NOT NULL,  
  
    password_hash TEXT,  
  
    phone_number VARCHAR(20),  
  
    role VARCHAR(20) DEFAULT 'user', -- 'user', 'admin', 'partner'  
  
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW(),  
  
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
  
);
```

-- 2. Push Notification Subscriptions

```
CREATE TABLE push_subscriptions (  
  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    user_id UUID REFERENCES users(id) ON DELETE SET NULL,  
  
    device_token TEXT UNIQUE NOT NULL,
```

```
browser VARCHAR(50),  
  
os VARCHAR(50),  
  
is_active BOOLEAN DEFAULT TRUE,  
  
subscribed_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
  
);
```

-- 3. Events Directory

```
CREATE TABLE events (  
  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
  title VARCHAR(255) NOT NULL,  
  
  slug VARCHAR(255) UNIQUE NOT NULL,  
  
  short_summary TEXT,  
  
  detailed_description TEXT,  
  
  banner_image_url TEXT,  
  
  devnovate_registration_url TEXT NOT NULL,  
  
  category VARCHAR(50) NOT NULL, -- 'hackathon', 'placement', 'workshop'  
  
  event_type VARCHAR(20) DEFAULT 'pan_india', -- 'virtual', 'onsite', 'hybrid'  
  
  location VARCHAR(150),  
  
  start_date TIMESTAMP WITH TIME ZONE NOT NULL,
```

```
end_date TIMESTAMP WITH TIME ZONE,  
  
is_featured BOOLEAN DEFAULT FALSE,  
  
is_published BOOLEAN DEFAULT TRUE,  
  
created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
  
);
```

-- 4. Ad Popups & Banners

```
CREATE TABLE ad_campaigns (  
  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    title VARCHAR(150) NOT NULL,  
  
    image_url TEXT NOT NULL,  
  
    destination_url TEXT NOT NULL,  
  
    target_page VARCHAR(100) DEFAULT '/events',  
  
    scroll_threshold_percent INT DEFAULT 50,  
  
    is_active BOOLEAN DEFAULT TRUE,  
  
    start_date TIMESTAMP WITH TIME ZONE,  
  
    end_date TIMESTAMP WITH TIME ZONE,  
  
    created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
  
);
```

-- 5. Service Inquiries / Leads

```
CREATE TABLE service_inquiries (  
  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
  client_name VARCHAR(120) NOT NULL,  
  
  organization_name VARCHAR(200) NOT NULL,  
  
  organization_type VARCHAR(50), -- 'college', 'corporate', 'startup'  
  
  email VARCHAR(255) NOT NULL,  
  
  phone VARCHAR(20),  
  
  service_type VARCHAR(100) NOT NULL,  
  
  budget_range VARCHAR(50),  
  
  message TEXT,  
  
  status VARCHAR(30) DEFAULT 'pending', -- 'pending', 'contacted', 'closed'  
  
  created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()  
  
);
```

7. Developer Sprint Execution Plan

Sprint 1: Core Setup & Static Routes

- Initialize Next.js repository with TypeScript, Tailwind CSS, and Shadcn UI.
- Implement responsive layout: Header, Navigation bar, and Footer across all viewports.
- Build static marketing views: `/` (Home), `/our-team`, and `/services`.
- Set up PostgreSQL schema with Prisma ORM.

Sprint 2: Events Directory & Redirection Engine

- Implement `/events` listing page with category and search filters.
- Integrate the Devnovate external redirect handler with UTM tracking.
- Build the scroll-depth detection hook (`useScrollTrigger`) and ad popup modal with session frequency capping.

Sprint 3: Authentication & Push Notifications

- Configure NextAuth for email/password and Google OAuth.
- Implement Service Worker registration for browser push notifications.
- Connect Firebase Cloud Messaging (FCM) / OneSignal and wire `/api/notifications/subscribe`.
- Create `/contact` and Service Lead submission API with automated email alerts.

Prepared for: Development Engineering Team

Status: Approved for Implementation